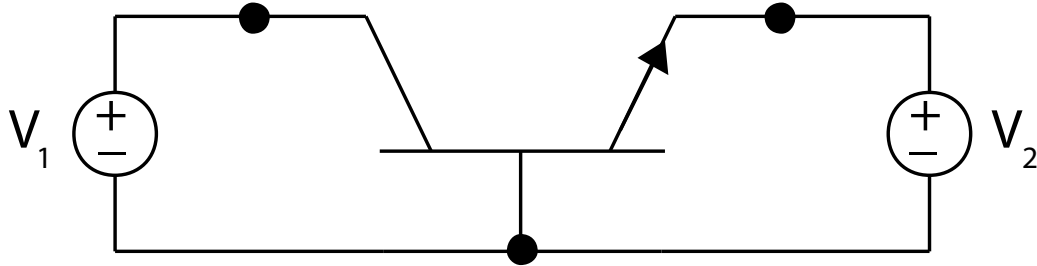


ECE 3040B PRACTICE TEST #2

1. (12 points)

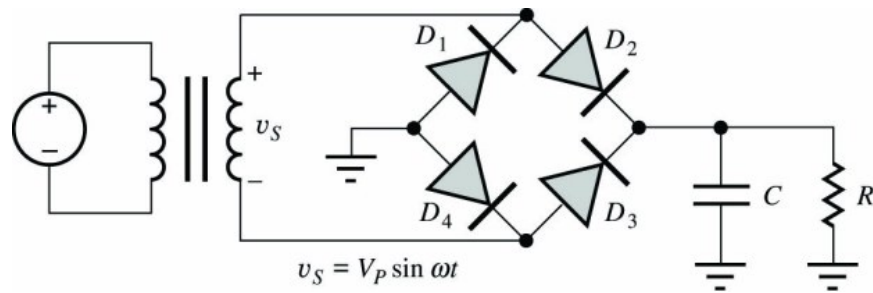


- (a) (2 points) What type of BJT is the transistor shown above? (NPN or PNP)
- (b) (2 points) Identify emitter, base and collector terminals and clearly show on the figure above.
- (c) (4 points) Determine the mode of operation under following conditions:

V_1	V_2	Mode of Operation
Positive	Positive	
Positive	Negative	
Negative	Positive	
Negative	Negative	

- (d) (4 points) Clearly show the **direction of current flow** at the emitter, base and collector terminals on the figure above when the transistor is operated in the **forward active mode**.

2. (6 points) In the rectifier circuit given below,



(a) (2 points) Which diode(s) conduct in negative cycles?

(b) (2 points) Turn-on voltage for each diode in the circuit is V_{on} . What is the output DC voltage?

(c) (2 points) How can you reduce the ripple voltage of this rectifier circuit?

3. **(4 points)** Why is the current gain different under forward active and inverted biasing of a BJT? Which of the two biasing conditions leads to higher current gain and why?
4. **(4 points)** Explain the base width modulation (Early effect) in a BJT? What small signal parameter does it affect?
5. **(4 points)** Explain the pinch-off point in a MOSFET. Under what conditions does it happen? You can support your explanation with a sketch.

6. **(10 points)** Sketch the energy band diagram and charge diagram of an ideal MOS structure for the following conditions. Also indicate the biasing regime for each case.

(a) (5 points) $\phi_F = -12kT/q$, $\phi_S = -26kT/q$

(b) (5 points) $\phi_F = 9kT/q$, $\phi_S = 6kT/q$

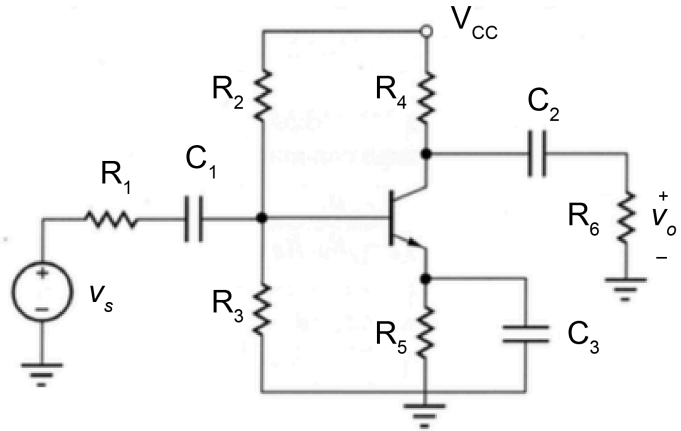
7. **(10 points)** Consider a NPN BJT with doping densities $N_E \gg N_B > N_C$.
- (a) (5 points) Sketch the energy band diagram under **forward active mode biasing**. Clearly indicate emitter, base and collector regions and label E_C , E_V , E_i and E_F .
- (b) (5 points) Roughly sketch the **total minority carrier** concentration in the emitter, base and collector regions under **forward active mode bias**. Please indicate the minority carrier type and equilibrium concentrations, mark the emitter, base and collector in your sketch, and assume $W/L_B \ll 1$.

Problem 1 (10 points): Consider a BJT with the doping concentrations for the emitter, base and collector are 10^{18} , 10^{16} , and $5 \cdot 10^{15} \text{ cm}^{-3}$, respectively. The base width is $2 \text{ }\mu\text{m}$, the diffusion lengths of the minority carriers are $L_E = L_C = 6 \text{ }\mu\text{m}$ and $L_B = 20 \text{ }\mu\text{m}$ and the diffusion coefficients are $D_E = D_B = D_C = 5 \text{ cm}^2/\text{s}$.

(a) (5 points) Calculate the emitter efficiency, the base transport factor, and the common emitter current gain of the transistor in **active** mode assuming $W \ll L_B$.

(b) (5 points) Calculate the emitter efficiency, the base transport factor, and the common emitter current gain of the transistor in **inverse** mode assuming $W \ll L_B$.

Problem 2 (40 points): For the circuit shown on the right, assume the transistor is operated in the active mode, $\beta_F = 100$, $V_{ON} = 0.7$ V, $V_A = 100$ V, $R_1 = 1$ k Ω , $R_2 = 48$ k Ω , $R_3 = 13$ k Ω , $R_4 = 8$ k Ω , $R_5 = 4$ k Ω , $R_6 = 150$ k Ω and $V_{CC} = 15$ V.



(a) (20 points) Calculate the Q-point, i.e. (I_C , V_{CE}), of the BJT. Hint: Draw the DC equivalent circuit (5 points). First find the Thevenin equivalent of the base bias circuit (5 points) and then calculate the transistor Q-point (10 points).

(b) (20 points) Calculate the signal source voltage gain v_o/v_s of the circuit. Hint: Draw the AC equivalent circuit and simplify the circuit by calculating its Thevenin equivalent (5 points). Calculate the small signal model parameters and replace the transistor with its small signal equivalent circuit (10 points). Then calculate the voltage gain of the circuit (5 points).

